

- 8 (a)** State what is meant by simple harmonic motion.

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 [2]

- (b)** A block of mass m is held on a smooth horizontal surface by means of two identical springs, each of spring constant k . The springs are attached to fixed points, as shown in Fig. 8.1.

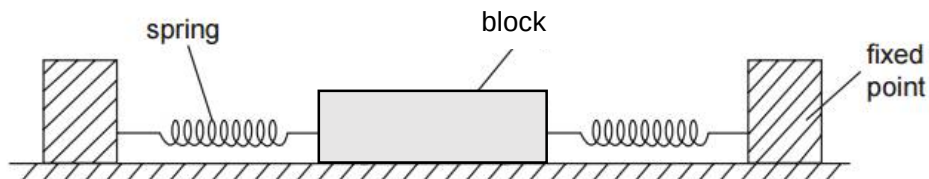


Fig. 8.1

When the block is in equilibrium, the extension of each spring is e .

The block is displaced a small distance x to the right along the axis of the springs. Both springs remain extended.

- (i)** The block is then released.

Show that the expression for the acceleration of the block is given by

$$a = -\frac{2k}{m}x .$$

[3]

- (ii) The mass of the block is 600 g and the spring constant of the springs is 50 N m^{-1} .

Determine the frequency of oscillation of the block.

frequency = Hz [2]

- (iii) If the block has a speed of 1.4 m s^{-1} when passing the equilibrium position, determine the amplitude of the oscillations.

amplitude = m [2]

- (iv) For the oscillations of the block, determine

1. the total energy E_T ,

$E_T = \dots\dots\dots \text{ J [2]}$

2. the displacement x at which the potential energy E_p and the kinetic energy E_k of the oscillations are equal.

$x = \dots\dots\dots \text{ m [2]}$

- (v) On Fig. 8.2, sketch graphs, with appropriate values, to show the variation with displacement x of

1. the total energy (label this line E_T),
2. the kinetic energy (label this line E_k),
3. the potential energy (label this line E_p).

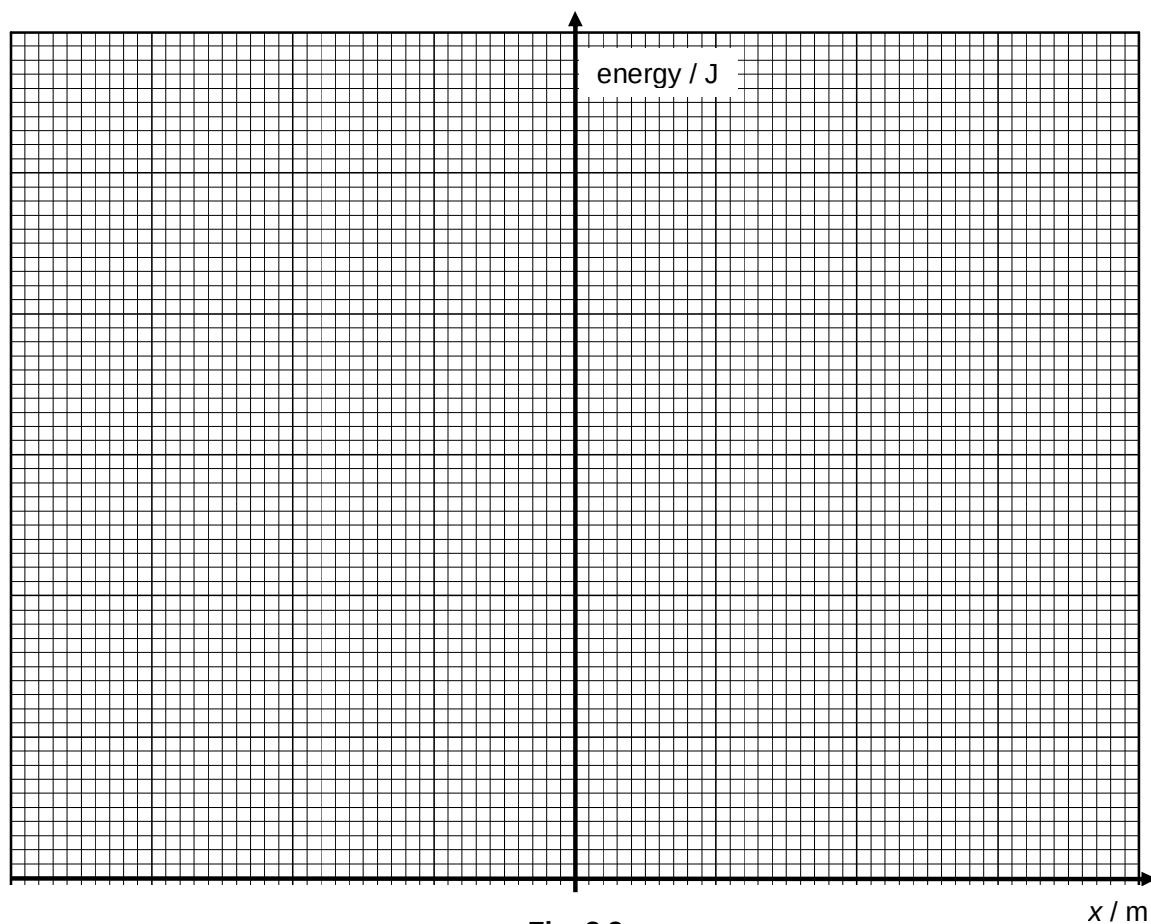


Fig. 8.2

 x / m

[3]

(vi) State and explain how the oscillation is affected by replacing

1. the block with one of larger mass,

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 [2]

2. the smooth surface with a rough one.

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 [2]

9 (a) Radioactive decay is a random and a spontaneous process